

Reference guide: Python functions for the discovery of a dataset

Python reference guide for EDA: Discovering

Use the following Python Pandas functions to help you learn about a dataset when you encounter it for the first time.

DataFrame.head(X)

- The head() function will display the number of dataset rows you input in the argument field.
- For the “X” in the argument field, input the number of rows you want displayed in a Python notebook. The default is 5 rows.
- Once executed, the head() function looks like this:

```
df.head(10)
```

n/a	Date	number of strikes	center point geom
0	2018-01-03	194	POINT(-75 27)
1	2018-01-03	41	POINT(-78.4 29)
2	2018-01-03	33	POINT(-73.9 27)
3	2018-01-03	38	POINT(-73.8 27)
4	2018-01-03	92	POINT(-79 28)
5	2018-01-03	119	POINT(-78 28)
6	2018-01-03	35	POINT(-79.3 28)
7	2018-01-03	60	POINT(-79.1 28)

8	2018-01-03	41	POINT(-78.7 28)
9	2018-01-03	119	POINT(-78.6 28)

Note: In a Python notebook, the results of `head()` will not include a table with visible grid lines.

DataFrame.info(X)

- The `info()` function will display a summary of the dataset, including the range index, dtypes, column headers, and memory usage.
- Leaving the argument field blank will return a full summary. As an option, in the argument field you can type in “`show_counts=True`,” which will not return any null fields.
- Once executed, the `info()` function looks like this:

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3401012 entries, 0 to 3401011
Data columns (total 3 columns):
#   Column          Dtype
--  ----
0   date            object
1   number_of_strikes  int64
2   center_point_geom  object
Dtypes: int64(1), object(2)
Memory usage 77.8+ MB
```

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Dataframe.describe(X)

- The `describe()` function will return descriptive statistics of the entire dataset, including total count, mean, minimum, maximum, dispersion, and distribution.

- Leaving the argument field blank will default to returning a summary of the data frame's statistics. As an option, you can use "include=[X]" and "exclude=[X]" which will limit the results to specific data types, depending on what you input in the brackets.
- Once executed, the describe() function looks like this:

```
df_joined.describe()
```

N/A	longitude	latitude	number_of_strikes_x	number_of_strikes_y
count	717530.00	717530.00	717530.00	323700.00000
mean	-90.875445	33.328572	21.637081	25.410587
std	13.648429	7.938831	48.02952	57.421824
min	-133.9000	16.600000	1.00000	1.000000
25%	-102.80000	26.900000	3.00000	3.000000
50%	-90.300000	33.200000	6.00000	8.000000
75%	-80.900000	39.400000	21.00000	24.000000
max	-43.800000	51.700000	2211.00000	2211.000000

Note: In a Python notebook, the results of describe() will not include a table with visible grid lines.

DataFrame.shape

- 'Shape' returns a tuple representing the dimensions of the dataset by number of rows and columns. The code will look something like this:

```
Df.shape
```

```
(3401012, 3)
```

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Key takeaways

`Head()`, `info()`, `describe()`, and `shape` are all Python functions that data scientists can use to understand a dataset at a high level. The information learned from running these functions will serve to inform the remainder of your EDA work when you use Python to analyze data throughout your career.

Resources for more information

For more information on the EDA discovering functions above and others like it, you can use the online Pandas reference guide:

- [A list of Pandas dataframe functions](#)